Calculus II

Solution to Assignment 3

Part I. Regular Problems

1. Determine the radius and interval of convergence of the following series:

   (a) $\sum_{k=1}^{\infty} \frac{(-1)^k}{k \cdot 3^k}(x-1)^k$
   Note that

$$\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty}\left|\frac{(-1)^{k+1}k3^k}{(-1)^k(k+1)3^{k+1}}\right| = \frac{1}{3}\lim_{k\to\infty}\frac{k}{k+1} = \frac{1}{3}$$

   So the radius of convergence is 3 and the series converges absolutely for $x \in (-2, 4)$. When $x = -2$,

$$\sum_{k=0}^{\infty}\frac{(-1)^k}{k3^k}(-3)^k = \sum_{k=0}^{\infty}\frac{1}{k}$$

   which is the divergent harmonic series.
   When $x = 4$,

$$\sum_{k=0}^{\infty}\frac{(-1)^k}{k\cdot 3^k}3^k = \sum_{k=0}^{\infty}\frac{(-1)^k}{k}$$

   which converges by the Alternating Series Test. So the interval of convergence is $(-2, 4]$ and $r = 3$.

   (b) $\sum_{k=1}^{\infty} \frac{(k!)^2}{(2k)!}x^{2k+1}$

Write $a_k = \frac{(k!)^2}{(2k)!}x^{2k+1}$. Using the Ratio Test, we have

$$\begin{aligned}\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| &= \lim_{k\to\infty}\left|\frac{(k+1)^2}{(2k+2)(2k+1)}x^2\right| \\ &= \left|\frac{1}{4}x^2\right| < 1\end{aligned}$$

for $-2 < x < 2$. So, the series converges for $-2 < x < 2$ and diverges for $x < -2$ or $x > 2$. Since the Ratio Test gives no conclusion at $x = -2$ and $x = 2$, we must test separately at $x = -2$ and $x = 2$.

$$\begin{aligned}a_k &= \frac{(k!)^2}{(2k)!}x^{2k+1} \\ &= \frac{(k!)^2}{(2k)!}2^{2k+1} \\ &= \frac{(k!)}{1.3\cdot 5\cdot 6\ldots.(2k-1)}2^{k+1} \\ &= \frac{(2\cdot 4\cdot 6\cdot 8\ldots.(2k))}{1\cdot 3\cdot 5\cdot 6\ldots(2k-1)}\cdot 2 > 4\end{aligned}$$

Since $\lim_{k\to\infty} a_k \neq 0$, the given series diverges for $x = 2$. Similarly the given series diverges at $x = -2$. Thus the power series converges at all x in the interval $(-2, 2)$ and diverges at all x outside of this interval. The radius of convergence is $r = 2$

(c) $\sum_{k=0}^{\infty} k!(x+1)^k$

Note that

$$\begin{aligned}\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| &= \lim_{k\to\infty}\left|\frac{(k+1)!}{k!}\right| \\ &= \lim_{k\to\infty}(k+1) = \infty\end{aligned}$$

so this series converges absolutely for $x = -1$ and $r = 0$.

(d) $\sum_{k=1}^{\infty}\frac{(-1)^k}{\sqrt{k}}(3x-1)^k$

We can write

$$\sum_{k=1}^{\infty}\frac{(-1)^k}{\sqrt{k}}(3x-1)^k = \sum_{k=1}^{\infty}\frac{(-1)^k\cdot 3^k}{\sqrt{k}}(x-\frac{1}{3})^k$$

and note that

$$\begin{aligned}\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| &= \lim_{k\to\infty}\left|\frac{3\sqrt{k}}{\sqrt{k+1}}\right| \\ &= 3\end{aligned}$$

So, the radius of convergence is $1/3$ and the series converges for $0 < x < \frac{2}{3}$ and diverges for $x < 0$ or $x > \frac{2}{3}$. Since the Ratio Test gives no conclusion at $x = 0$ and $x = \frac{2}{3}$, we must test separately at $x = 0$ and $x = \frac{2}{3}$.

For $x = 0$

$$\sum_{k=1}^{\infty} \frac{(-1)^k(-1)^k}{\sqrt{k}} = \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$$

Hence the series diverges. Therefore, the given series diverges for $x = 0$. For $x = \frac{2}{3}$, $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}$ is alternative series and $\lim_{k\to\infty} \frac{1}{\sqrt{k}} = 0$. Hence the series converges. Therefore, the given series converges for $x = \frac{2}{3}$. Thus the power series converges at all x in the interval $\left(0, \frac{2}{3}\right]$ and diverges at all x outside of this interval and the radius of convernce is

$$r = \frac{1}{3}$$

(e) $\sum_{k=2}^{\infty}(k+3)^2(2x-3)^k$
We write

$$\sum_{k=2}^{\infty}(k+3)^2(2x-3)^k = \sum_{k=2}^{\infty} 2^k \cdot (k+3)^2(x-\frac{3}{2})^k$$

and note that

$$\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty}\left|\frac{2(k+4)^2}{(k+3)^2}\right| = |\lim_{k\to\infty} 2\left(\frac{k+4}{k+3}\right)^2 = 2$$

so the radius of convergence is $1/2$ and the series converges absolutely for $x \in (1, 2)$ If $x = 1, \sum_{k=2}^{\infty}(k+3)^2(-1)^k$ and $\lim_{k\to\infty}(k+3)^2 = \infty$, so the series diverges by the divergence test. If $x = 2, \sum_{k=2}^{\infty}(k+3)^2(1)^k$ and $\lim_{k\to\infty}(k+3)^2 = \infty$, so the series diverges by the divergence est. The interval of divergence is $(1, 2)$.

2. Find a power series representation of $f(x)$ about $c = 0$. Also, determine the radius and interval of convergence.

(a) $f(x) = \frac{2}{1-x}$
Note that

$$f(x) = \frac{2}{1-x} = 2\sum_{k=0}^{\infty} x^k = \sum_{k=0}^{\infty} 2x^k$$

This is a geometric series that converges when $|x| < 1$, so the interval of convergence is $(-1, 1)$ and $r = 1$.

(b) $f(x) = \frac{2}{1-x^2}$ Note that

$$\begin{aligned} f(x) = \frac{2}{1-x^2} &= 2\left(\frac{1}{1-(x^2)}\right) \\ &= 2\sum_{k=0}^{\infty}\left(x^2\right)^k = \sum_{k=0}^{\infty} 2x^{2k} \end{aligned}$$

and there is a geometric series involved that converges when $\left|x^2\right| < 1$, so the interval of convergence is $(-1, 1)$ and $r = 1$.

(c) $f(x) = \frac{3x}{1+x^2}$
Note that

$$\begin{aligned} f(x) &= \frac{3x}{1+x^2} = 3x\left(\frac{1}{1-(-x^2)}\right) \\ &= 3x\sum_{k=0}^{\infty}\left(-x^2\right)^k = 3x\sum_{k=0}^{\infty}(-1)^k x^{2k} \\ &= \sum_{k=0}^{\infty}(-1)^k 3x^{2k+1} \end{aligned}$$

and for the involved geometric series to converge, we need to require that $\left|-x^2\right| < 1$, so the interval of convergence is $(-1, 1)$ and $r = 1$.

3. Find a power series representation and radius of convergence by integrating or differentiating one of the series from Q2.

(a) $f(x) = 2\ln(1-x)$
We have seen that $\frac{2}{1-x} = \sum_{k=0}^{\infty} 2x^k$ for $x \in (-1, 1)$. Since

$$\begin{aligned} \int \frac{2dx}{1-x} &= \sum_{k=0}^{\infty}\int 2x^k dx \\ &= C_1 + \sum_{k=0}^{\infty}\frac{2x^{k+1}}{k+1} = C_1 + \sum_{k=1}^{\infty}\frac{2x^k}{k} \end{aligned}$$

and

$$\int \frac{2dx}{1-x} = -2\ln(1-x) + C_2$$

we obtain $2\ln(1-x) = C_3 - \sum_{k=1}^{\infty}\frac{2}{k}x^k$. Substituting $x = 0$, we find $C_3 = 0$, so

$$2\ln(1-x) = -\sum_{k=1}^{\infty}\frac{2}{k}x^k$$

for $x \in (-1, 1)$.

(b) $f(x) = \frac{2x}{(1-x^2)^2}$
We have seen that $\frac{2}{1-x^2} = \sum_{k=0}^{\infty} 2x^{2k}$ with $r = 1$. Taking the derivative of both sides gives

$$\frac{d}{dx}\left(\frac{2}{1-x^2}\right) = \sum_{k=1}^{\infty} 2\frac{d}{dx}x^{2k}$$
$$\frac{4x}{(1-x^2)^2} = \sum_{k=1}^{\infty} 2 \cdot 2kx^{2k-1}$$
$$\frac{2x}{(1-x^2)^2} = \sum_{k=1}^{\infty} 2kx^{2k-1} \text{ with } r = 1$$

(c) $f(x) = \ln\left(1+x^2\right)$
We have seen that $\frac{3x}{1+x^2} = \sum_{k=0}^{\infty}(-1)^k 3x^{2k+1}$ with $r = 1$. Integrating both sides gives

$$\int \frac{3x}{1+x^2}dx = C + \sum_{k=0}^{\infty}(-1)^k 3\int x^{2k+1}dx$$
$$\frac{3}{2}\ln\left(1+x^2\right) = \sum_{k=0}^{\infty}\frac{(-1)^k 3x^{2k+2}}{2k+2} + C$$
$$\ln\left(1+x^2\right) = \sum_{k=0}^{\infty}\frac{(-1)^k x^{2k+2}}{k+1} + C$$

Taking $x = 0$, $\ln(1) = \sum_{k=0}^{\infty}\frac{(-1)^k(0)^{2k+2}}{k+1} + C = C$ so that $C = \ln(1) = 0$.
Thus, $\ln\left(1+x^2\right) = \sum_{k=0}^{\infty}\frac{(-1)^k x^{2k+2}}{k+1}$ with $r = 1$.

Part II. Group Problems

1. Find the radius of convergence and interval of convergence of the power series.

(a) $\sum_{n=1}^{\infty}(-1)^n nx^n$
We have

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(-1)^{n+1}(n+1)}{(-1)^n n}\right| == \lim_{n\to\infty}\left(1+\frac{1}{n}\right) = 1.$$

So the radius of convergence is 1 and the series $\sum_{n=1}^{\infty}(-1)^n nx^n$ converges when $|x| < 1$, so the radius of convergence. Now we'll check the endpoints, that is, $x = \pm 1$. Both series $\sum_{n=1}^{\infty}(-1)^n n(\pm 1)^n = \sum_{n=1}^{\infty}(\mp 1)^n n$ diverge by the Test for Divergence since $\lim_{n\to\infty}|(\mp 1)^n n| = \infty$. Thus, the interval of convergence is $I = (-1, 1)$.

(b) $\sum_{n=2}^{\infty} \frac{5^n}{n} x^n$
We have

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{5^{n+1}}{(n+1)} \cdot \frac{n}{5^n}\right| = \lim_{n\to\infty} \left|\frac{5n}{n+1}\right| = \lim_{n\to\infty} \left(\frac{5}{1+1/n}\right) = 5.$$

So the radius of convergence is $1/5$ and the series $\sum_{n=2}^{\infty} \frac{5^n}{n} x^n$ converges when $|x| < \frac{1}{5}$. When $x = \frac{1}{5}$, the series $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges since it is the (partial) harmonic series. When $x = -\frac{1}{5}$, the series $\sum_{n=2}^{\infty} \frac{(-1)^n}{n}$ converges by the Alternating Series Test. Thus, the interval of convergence is $\left[-\frac{1}{5}, \frac{1}{5}\right)$.

(c) $\sum_{n=1}^{\infty} \frac{x^{2n}}{n!}$
If $a_n = \frac{x^{2n}}{n!}$, then $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{x^{2n+2}}{(n+1)!} \cdot \frac{n!}{x^{2n}}\right| = \lim_{n\to\infty} \frac{\left|x^2\right|}{n+1} = 0 < 1$ for all real x. So, by the Ratio Test, $R = \infty$ and $I = (-\infty, \infty)$.

2. (a) i. $f(x) = \frac{3}{1+x^2}$
Note that

$$f(x) = \frac{3}{1+x^2} = 3\left[\frac{1}{1-(-x^2)}\right]$$
$$= 3\sum_{k=0}^{\infty} \left(-x^2\right)^k = \sum_{k=0}^{\infty} 3(-1)^k x^{2k}$$

This involves a geometric series that converges when $\left|x^2\right| < 1$, so the interval of convergence is $(-1, 1)$ and $r = 1$.

ii. $f(x) = \frac{3}{x-1}$
Note that

$$f(x) = \frac{3}{x-1} = -3\left(\frac{1}{1-x}\right)$$
$$= -3\sum_{k=0}^{\infty} x^k = \sum_{k=0}^{\infty} (-3)x^k.$$

This involves a geometric series that converges when $|x| < 1$, so the interval of convergence is $(-1, 1)$ and $r = 1$.

(b) i. $f(x) = 3\tan^{-1} x$
We have seen that

$$\frac{3}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k 3x^{2k} \text{ with } r = 1$$

Integrating both sides gives

$$\int \frac{3}{1+x^2}dx = \sum_{k=0}^{\infty} 3(-1)^k \int x^{2k}dx$$

$$3\tan^{-1} x = \sum_{k=0}^{\infty} \frac{3(-1)^k x^{2k+1}}{2k+1} + c$$

Taking $x = 0$ we get $3\tan^{-1} 0 = \sum_{k=0}^{\infty} \frac{3(-1)^k (0)^{2k+1}}{2k+1} + c = c$ so that $c = 3\tan^{-1}(0) = 0$. Thus

$$3\tan^{-1} x = \sum_{k=0}^{\infty} \frac{3(-1)^k x^{2k+1}}{2k+1}$$

with $r = 1$.

ii. We have seen that $\frac{3}{x-1} = -3\sum_{k=0}^{\infty} x^k$ for $x \in (0, 2)$. Since

$$\frac{d}{dx}\left(\frac{3}{x-1}\right) = -3\sum_{k=0}^{\infty} \frac{d}{dx}x^k$$

$$= -3\sum_{k=1}^{\infty} kx^{k-1} = -3\sum_{k=0}^{\infty}(k+1)x^k, \text{ and}$$

$$\frac{d}{dx}\left(\frac{3}{x-1}\right) = -\frac{3}{(x-1)^2}, \text{ we obtain}$$

$$\frac{3}{(x-1)^2} = 3\sum_{k=0}^{\infty}(k+1)x^k \text{ for } x \in (0, 2)$$